



Numerical investigation of mixed convection heat transfer of water- Fe_3O_4 nanofluid in a corrugated channel with moving walls

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Received: 13 February 2026 / Accepted: 16 August 2026
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Abstract

Increasing heat transfer in heat exchangers and ducts filled with various operating fluids for heat transfer is one of the main concerns of researchers and engineers in the energy conversion in the field of thermal engineering. Two examples of methods for increasing heat transfer in these ducts are the use of nanofluids instead of conventional fluids and the use of vortex generators in the fluid path or in the duct walls. In the present numerical study, the thermal performance of a water- Fe_3O_4 nanofluid is evaluated inside a two-dimensional channel with sinusoidal corrugated walls designed to promote vortex generation. In this regard, a steady heat flux from one and two walls was applied to the fluid and the wall was considered in two states, stationary and moving, and the effect of wall velocity, nanofluid mass flow rate, and transition time on the thermal parameters of the fluid between the two walls was investigated. Among the outstanding results obtained in this study, it can be noted that at lower velocities (between 0.01 and 0.1 m s^{-1}), the temperature distribution in the channel is wider, meaning that more areas of the fluid inside the channel experience a temperature increase. The maximum temperature in the channel occurs in parts of it where the moving wall exits it, and increasing the velocity beyond 0.5 m s^{-1} no longer has a significant effect on the maximum temperature generated in the nanofluid. Overall, the findings indicate that adjusting the moving wall velocity alongside the nanoparticle concentration can considerably improve the thermal management of corrugated channels.

Keywords Mixed convection · Water- Fe_3O_4 nanofluid · Corrugated channel · Moving wall · Heat transfer enhancement · Flow hydrodynamics

List of symbols

c	Specific heat of nanofluid ($\text{J}(\text{kg} \cdot \text{K})^{-1}$)	j_T	Thermophoretic mass flux ($\text{kg}(\text{m}^2 \cdot \text{s})^{-1}$)
c_p	Specific heat of nanoparticle ($\text{J}(\text{kg} \cdot \text{K})^{-1}$)	j_p	Total nanoparticle mass flux ($\text{kg}(\text{m}^2 \cdot \text{s})^{-1}$)
D_B	Brownian diffusion coefficient (m^2s^{-1})	k	Thermal conductivity ($\text{W}(\text{m} \cdot \text{K})^{-1}$)
D_T	Thermophoretic diffusion coefficient (m^2s^{-1})	k_B	Boltzmann constant (J K^{-1})
d_p	Nanoparticle diameter (m)	l	Characteristic length (e.g., plate length)(m)
h_p	Enthalpy of nanoparticle (J kg^{-1})	$\dot{M}$	Mass production rate ($\text{kg}(\text{m}^3 \cdot \text{s})^{-1}$)
j_B	Brownian mass flux ($\text{kg}(\text{m}^2 \cdot \text{s})^{-1}$)	P	Pressure (Pa)
		$\vec{q}$	Heat flux vector (W m^{-2})
		q''	Applied wall heat flux (W m^{-2})
		T	Temperature (K)
		t	Time (s)
		U	Free-stream velocity (outside boundary layer)(m s^{-1})
		u	Velocity component in x-direction (m s^{-1})
		v	Velocity component in y-direction (m s^{-1})
		$\vec{v}$	Velocity vector (m s^{-1})
		w	Channel width or geometric parameter (m)
		x	Streamwise coordinate (m)
		y	Wall-normal coordinate (m)

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ρ	Density (kg m^{-3})
τ	Stress tensor (Pa)

Greek symbols

δ	Boundary layer thickness (m)
φ	Nanoparticle volume fraction
μ	Dynamic viscosity ($\text{Pa} \cdot \text{s}$)

Subscripts

f	Base fluid
nf	Nanofluid
p	Nanoparticle

Abbreviations

CNT	Carbon nanotube
MHD	Magnetohydrodynamics
TiO_2	Titanium dioxide
Al_2O_3	Aluminum oxide
SiO_2	Silicon dioxide
CuO	Copper(II) oxide
Fe_3O_4	Magnetite (Iron(II,III) Oxide)

Introduction

The design of compact heat exchangers requires continuous improvement in thermal performance. The use of complex geometries, such as corrugated or sinusoidal channels, is a proven passive technique to enhance heat transfer. These geometries disrupt the development of the thermal boundary layer and promote flow mixing. However, this enhancement often comes at the cost of increased pressure drop, making optimization of channel shape and flow hydrodynamics a critical area of research. Furthermore, the inclusion of moving walls can further modify the shear layers and temperature distribution, providing active control over convective heat transfer. The process of convection is usually a thermal interaction between a surface and a fluid in contact, which may be at rest or in motion, as in the case of fluid in pipes. Convection can occur not only in a uniform fluid, but also during the mixing of fluids or layers of different temperatures. This heat transfer mechanism plays a vital role in a wide range of engineering systems. Key applications include thermal power plants, heat exchangers used for heating or cooling, long pipelines for transporting oil and gas, and food processing and pasteurization systems. In the design, optimization, and management of these systems, it is essential to have information about how the heat transfer coefficient is mechanically distributed over internal surfaces, such as the inner walls of pipes and channels. This information allows engineers to accurately analyze the thermal performance of the system, increase energy efficiency, and prevent problems such as thermal stress concentrations or fouling. With these considerations in mind, conducting basic studies and fundamental research on all heat

transfer methods, especially convective heat transfer, is not only a scientific necessity but also a valuable and important investment for the industry sector. Adding depth to this field will pave the way for the development of more efficient and sustainable technologies in the future. One of the methods of increasing heat transfer is the addition of conductive solid nanoparticles to the base fluid, which improves the thermal conductivity of the base fluid (water, propylene glycol, and ethylene glycol). The term nanofluid was first used by Choi [1]. Nanofluids generally have improved thermophysical properties, including higher thermal conductivity and viscosity [2–4]. Researchers have developed various types of nanofluids using various nanoparticles such as TiO_2 [5, 6], Al_2O_3 [7, 8], SiO_2 [9, 10], CNT [11, 12], CuO [13, 14], and Fe_3O_4 [15, 16] and various base fluids, and have investigated their heat transfer mechanisms and thermophysical properties in more detail. Many studies have confirmed the improvement in thermal properties of nanofluids compared to base fluids. The superior thermal behavior of nanofluids is not simply due to the higher static thermal conductivity of solid nanoparticles. This behavior is also strongly influenced by physical phenomena at the microscale. Brownian motion of nanoparticles enhances microconvection and energy transfer in the base fluid, while thermophoresis causes particle migration due to temperature gradients and affects local concentration and thermal properties. Although Sort (thermal infiltration) and Dufour (thermal infiltration) effects can also affect the combined heat and mass transfer in multicomponent flows, the dominant mechanisms in standard single-phase nanofluid modeling are often controlled by effective properties modified by Brownian and thermophoretic mechanisms. Early studies showed that the addition of nanotubes to oil improved thermal conductivity [17]. Subsequent studies on nanofluids containing $\gamma\text{-Al}_2\text{O}_3$ nanoparticles in deionized water recorded a significant increase in convective heat transfer in laminar flow [18]. A special experimental system was designed to investigate these properties in more detail and a new relationship was proposed for the analysis of heat transfer data of nanofluids [19]. Numerical studies on two different configurations (uniformly heated tube and parallel disks) showed that the addition of nanoparticles to the base fluids significantly increased the heat transfer coefficient, and this improvement was intensified with increasing nanoparticle concentration [20]. One study emphasizes that the properties of nanofluids in the boundary layer can change under the influence of temperature gradients and thermophoresis, which leads to a decrease in viscosity and, consequently, an increase in heat transfer [21]. Numerous additional studies have explored the effects of nanofluids. Akbarinia and Behzadmehr performed a numerical analysis of laminar mixed convection for a water- Al_2O_3 nanofluid in a horizontal curved tube [22]. In a study conducted by Ho et al., the effects of uncertainties associated with two important parameters of nanofluids (thermal conductivity and effective dynamic

viscosity) on heat transfer due to natural convection in a laminar state, inside a chamber with a square cross section, were investigated. [23]. Mirmasoumi and Behzadmehr also conducted a numerical study on laminar mixed convection of water-Al₂O₃ nanofluid in a horizontal tube [24]. In another numerical study, Izadi et al. examined laminar forced convection of Al₂O₃-water nanofluid [25]. Raju et al. used computational fluid dynamics to study noise reduction in hydrofoils [26]. Additionally, Kuznetsov and Nield advanced a theoretical framework for nanofluid convection incorporating double-diffusion effects in porous media [27]. The study by Mokmeli and Saffar-Avval shows that the “dispersion” model explains the enhanced heat transfer in nanofluids better than traditional models due to the consideration of the irregular motion of nanoparticles. This model agrees well with the experimental data [28]. Sajjadi and Kazemi found that adding small amounts of titanium dioxide nanoparticles to water (less than 0.25%) significantly increased heat transfer, but further increases in concentration had no noticeable effect. The nanofluid also showed a higher pressure drop than pure water, which increased with increasing concentration. [29]. Numerical investigation by Bianco et al. shows that the heat transfer coefficient in the turbulent flow of water-Al₂O₃ nanofluid in a tube increases compared to the base fluid, using both single-phase and two-phase models.[30]. Numerical investigation by Manca et al. shows that water-Al₂O₃ nanofluids in ribbed channels improve heat transfer, but this improvement is associated with an increase in pumping power. The aim of the study was to find the optimal geometry for the balance between heat transfer and pressure drop. [31]. Yadav et al.’s study using linear stability analysis shows that increasing the thermal conductivity and viscosity of the nanofluid in a porous layer delays the onset of the dual-diffusion convection phenomenon.[32]. Göktepe et al. developed a macroscopic model to describe the hydrodynamic and thermal behavior of nanofluid flows in the entrance region of a uniformly heated tube [33]. Yadav et al. also investigated the influence of a magnetic field on the onset of convection in a uniformly internally heated nanofluid layer [34]. In a separate study, they analyzed the combined effect of temperature-dependent properties and nanoparticle concentration on double-diffusive convection in a rotating porous layer using novel boundary conditions for nanoparticle distribution [35]. Sheikholeslami et al. numerically simulated the impact of an external magnetic source on two-dimensional Fe₃O₄-water nanofluid flow [36]. Sheikholeslami and Shehzad studied the flow of water-copper oxide nanofluid in a porous semi-ring and showed that increasing the Hartmann number and adding nanoparticles improves heat transfer, while increasing buoyancy forces has the opposite effect [37]. Vo et al. employed a two-phase mixture model to examine the influence of nanoparticle shape on fluid flow and heat transfer characteristics of γ -AlOOH nanofluid in a sinusoidal wave channel [38].

Hosseini et al. evaluated the entropy generation of Al₂O₃-water nanofluid within a symmetrically heated horizontal porous microchannel [39]. Baslem et al. developed a heat transfer model for a longitudinal permeable fin fully wetted by nanofluid, incorporating natural convection and radiation effects [40]. Selimefendigil and Öztöp found that channel bifurcation and magnetic field slow down and increase the rate of phase change, respectively, while nanoparticles and increased Reynolds number accelerate the process. [41]. Ali’s experiments showed that increasing the concentration of SiO₂/water nanofluid to 0.007% improved heat transfer by 27%, which is due to increased conductivity and particle movement. [42]. Ali et al. examined the impacts of electromagnetic forces, combined free and forced convection, and resistive heating on ethylene glycol-based nanofluids flowing past a slender vertical needle [43]. Saleh et al. presented a study for effective thermal management of a channel with elastic sinusoidal barriers and multiple pulsating nanojet impacts [44]. Their results showed that the cooling efficiency is improved by 24.383% using pulsating flow. In another study by Saleh et al., a new heat sink with convex segment fins was designed to investigate the best heat dissipation methods to achieve processor cooling efficiency [45]. The results of the aforementioned study showed that using a large number of fins improved cooling efficiency by up to 74%.

Studies have also been conducted in the field of channels with movable walls [46, 47]. Nwaigwe et al., investigated the effects of moving channel wall, thermal radiation and variable thermal conductivity on the flow of a non-Newtonian fluid in a porous channel [48]. Pattnaik et al., analyzed the influence of buoyant forces of a conducting time-dependent nanofluid flow through porous moving walls [49]. Jeong and colleagues investigated the effects of liquid–solid wettability, wall column height, moving wall velocity, and channel height on the Couette flow in a nanochannel [50]. The results showed that the wall slip velocity decreases with increasing wettability due to the increase in the attractive force between the fluid and the solid wall.

To the best of the author’s knowledge, while many studies have investigated corrugated channels or moving walls separately, the novelty of the present work lies in the simultaneous investigation of the mixed convective heat transfer of water-Fe₃O₄Fe₃O₄ nanofluid in a channel with sinusoidal and moving walls. Furthermore, this study comprehensively analyzes the hydrodynamic and thermal behaviors in both steady and transient states, considering different boundary conditions such as heat flux on one or two top/bottom walls. This particular configuration provides useful information on heat transfer enhancement in complex geometries that can be used in the optimization of modern compact heat exchangers.

Table 1 Thermophysical properties of water, and Fe_3O_4 nanoparticles [36]

	$\rho / \text{kg m}^{-3}$	$C_p / \text{J kg}^{-1} \text{K}^{-1}$	$k / \text{W m}^{-1} \text{K}^{-1}$	$\mu / \text{Pa.s}$	dp / nm
Pure water	997.1	4179	0.613	0.001003	–
Fe_3O_4	5200	670	6	–	47

Formulation

Nanofluid properties

Assuming water as the base fluid in the simulation, the relevant thermophysical properties are presented in Table 1. This table contains the exact values of key properties including thermal conductivity, specific heat capacity, density and viscosity for both the base fluid (water) and the iron oxide nanoparticles [36].

The properties of nanofluids required for analysis are density, specific heat capacity, thermal conductivity, and viscosity. The following relationships can be used to calculate these properties [36]:

$$\rho_{\text{nf}} = (1 - \varphi)\rho_f + \varphi\rho_p \quad (1)$$

$$(\rho C_p)_{\text{nf}} = (1 - \varphi)(\rho C_p)_f + \varphi(\rho C_p)_p \quad (2)$$

$$k_{\text{nf}} = k_f \left[\frac{k_p + 2k_f - 2\varphi(k_f - k_p)}{k_p + 2k_f + \varphi(k_f - 2k_p)} \right] \quad (3)$$

$$\mu_{\text{nf}} = \frac{\mu_f}{1 - \varphi^{2.5}} \quad (4)$$

Continuity equations

In the present study, the thermal and flow behavior of the nanofluid is analyzed through numerical simulation. Therefore, the governing equations are introduced in their general conservation form, and detailed mathematical derivations are not presented. These equations correspond to the fundamental principles of mass, momentum,

and energy conservation that describe fluid flow and heat transfer.

Since the numerical solution is carried out using a computational solver, the objective of this section is to present the set of equations implemented in the model rather than to reproduce their theoretical derivation. The complete derivation of these equations is well established in the literature and can be found in standard references on fluid mechanics and heat transfer.

Accordingly, the continuity, momentum, and energy equations are employed to describe the flow and thermal fields in the channel, while the effective thermophysical properties of the nanofluid are incorporated to account for the influence of nanoparticle concentration on the heat transfer characteristics (Table 2).

To provide a generalized description of the flow behavior, the governing equations are expressed using dimensionless variables based on the characteristic length and velocity scales of the problem. The channel height H is selected as the characteristic length, while the inlet velocity U_{in} is used as the velocity scale. Accordingly, the dimensionless variables can be written as

$$X = \frac{x}{H}, Y = \frac{y}{H}, Z = \frac{z}{H}, U = \frac{u}{U_{\text{in}}}, V = \frac{v}{U_{\text{in}}}, \text{ and } P = p / (\rho U_{\text{in}}^2)$$

Under this formulation, the flow behavior is primarily characterized by the Reynolds number.

$Re = \frac{\rho U_{\text{in}} H}{\mu}$ which represents the ratio of inertial to viscous forces. Since the thermophysical properties of the nanofluid are assumed constant and the study focuses on forced convection conditions, the Reynolds number becomes the principal controlling parameter governing the flow and heat transfer characteristics. Therefore, the analysis in the present work is conducted by examining the effect of Reynolds number variations on the thermal and flow fields within the channel.

Table 2 Summarized equations for nanofluids

Fluid continuity equation	$\nabla \cdot \vec{v} = 0$
Nanoparticle continuity equation	$\frac{\partial \varphi}{\partial t} + \vec{v} \cdot \nabla \varphi = \nabla \cdot \left(D_B \nabla \varphi + D_T \frac{\nabla T}{T} \right)$ equation
Navier—Stokes	$\rho \left(\frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot \nabla \vec{v} \right) = -\nabla P - \nabla \cdot \mu \left[\nabla \vec{v} + (\nabla \vec{v})^T \right]$
Energy equation	$\rho c \frac{DT}{Dt} = \nabla \cdot (k \nabla T) - \rho_p c_p D_B \nabla \varphi \cdot \nabla T + \rho_p c_p D_T \frac{\nabla T \cdot \nabla T}{T}$

Fig. 1 Geometry studied in the present study

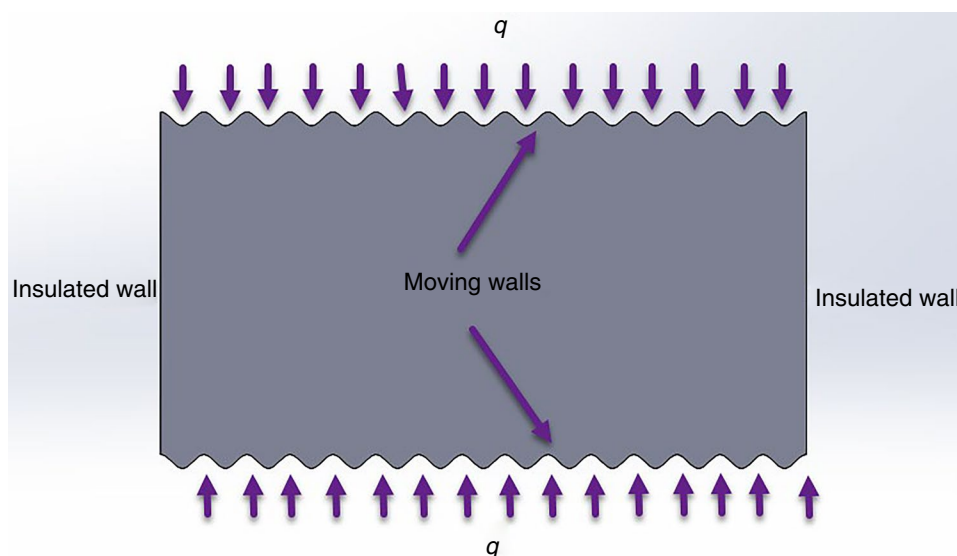


Table 3 Average temperature and errors in different size of elements

Element size (mm)	Quadrilateral elements		Rectangular elements	
	T_{mean}	Error	T_{mean}	Error
0.2	304.989		0.2	303.425
0.4	305.859	0.285256	0.4	303.452
0.6	304.328	0.500557	0.6	303.443
0.8	305.598	0.417313	0.8	303.461
1	303.736	0.609297	1	303.37
1.2	304.206	0.15474	1.2	303.47
1.4	306.329	0.697882	1.4	304.898
1.6	305.842	0.158979	1.6	304.12
1.8	306.329	0.159233	1.8	304.645
2	307.269	0.30686	2	305.25

Geometry and boundary conditions of the present study

The geometry considered for the analysis of the present problem is a two-dimensional duct with a length of 1000 mm and a width of 500 mm with sinusoidal walls where the amplitude of each wave is 20 mm which is shown in Fig. 1. To perform the analysis in the static wall mode, since the fluid is assumed to be stationary inside the chamber, the Reynolds number is zero and the side walls of the chamber are considered to be

insulating, and since in this case, the analysis is time-dependent, the initial temperature of the nanofluid is considered to be 300 K (Table 3).

Also, the analysis in the moving wall mode is considered to be time-independent (steady state) and, given the motion velocities used to move the wall, the flow will be turbulent. (Refer to Table 4.)

In the present numerical study, several simplifying assumptions are adopted to ensure a clear and tractable formulation of the governing equations while maintaining sufficient physical accuracy. The flow is considered incompressible and laminar, consistent with the operating conditions and the Reynolds number range investigated. The working fluid is treated as a Newtonian, single-phase, and homogeneous nanofluid, where the effective thermophysical properties represent the combined behavior of the base fluid and suspended nanoparticles. These properties are assumed constant, except where temperature-dependent variations are essential for capturing the thermal behavior.

Since the analysis focuses on forced convection under moderate temperature differences, radiation heat transfer and viscous dissipation effects are neglected. The thermophoretic influence associated with temperature gradients in the boundary layer is considered relevant to the physical interpretation of nanofluid behavior, although no slip-velocity model is employed. Under these assumptions, the continuity, momentum, and energy equations can be expressed in their general

Table 4 Important geometric and boundary parameters

Parameter	L_1 /(mm)	L_2 /(mm)	p/D	w/D	e/D	q''/W (m ² K) ⁻¹	T /(K)
Value	300	10000	0.03826	0.3	0.6	100	178

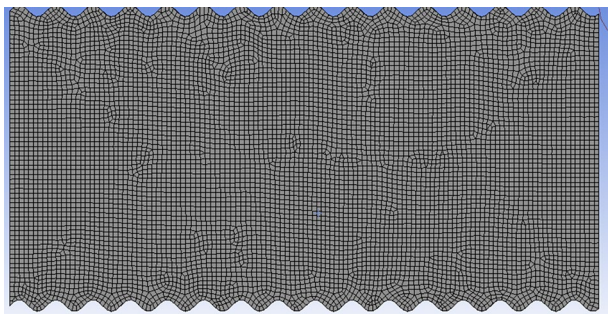


Fig. 2 Quadrilateral elements

conservation form and solved numerically to characterize the flow and heat transfer phenomena within the corrugated channel.

Discretization and mesh independence

Figure 2 shows the quadrilateral elements and Fig. 3 illustrate the triangular elements applied to the problem geometry. In order to obtain the appropriate element type and size for the analysis, a mesh convergence process was performed and the average temperature parameter of the enclosure was considered as the output for a steady state and a wall velocity of 0.2 m s^{-1} and a heat flux of 1000 W . The graphs related to this process are shown in the figure.

According to the diagram presented in Fig. 4, it can be seen that by changing the dimensions of the quadrilateral elements and reducing them, the outlet temperature from the duct, which is considered as a criterion, fluctuates regularly, so that the answers obtained first decrease with an oscillating state and then increase again. This shows that the answer obtained using cubic elements never converges to a specific value, so the answers obtained will be highly dependent on the solution grid. Therefore, it can be stated that the quadrilateral elements do not provide a suitable answer for the present problem and the answers obtained will never be independent of the solution grid.

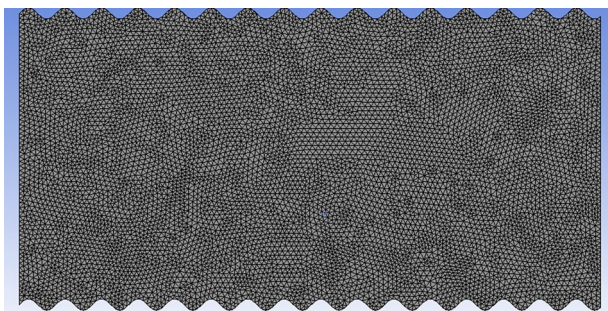


Fig. 3 Triangular elements

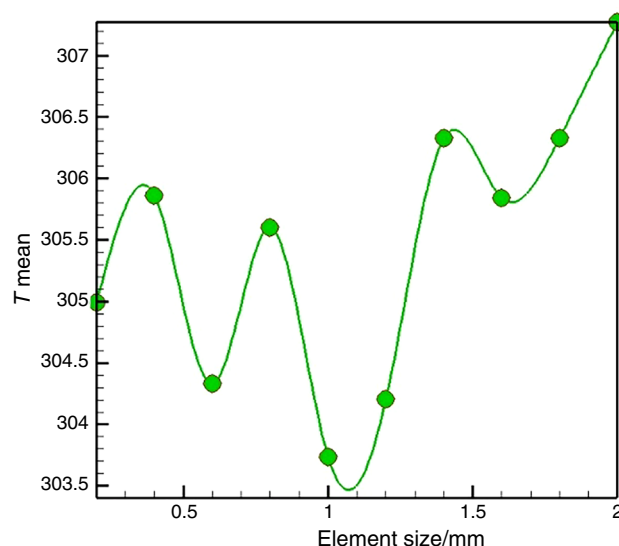


Fig. 4 Changes in the average temperature from the duct by changing the size of the quadrilateral elements

However, in Fig. 5, it can be seen that by reducing the dimensions of the triangular elements, the graph related to the outlet temperature from the duct shows a downward trend with an oscillation that ultimately leads to a horizontal line. This shows that the answers obtained by triangular elements in the dimensional range of less than 1 mm have solution independence from the grid and are so-called mesh convergence. Therefore, in order to continue the analysis of this problem, triangular elements with dimensions between 0.5 and 1 mm will be used. However, it should be noted that by reducing the dimensions of the elements, the analysis

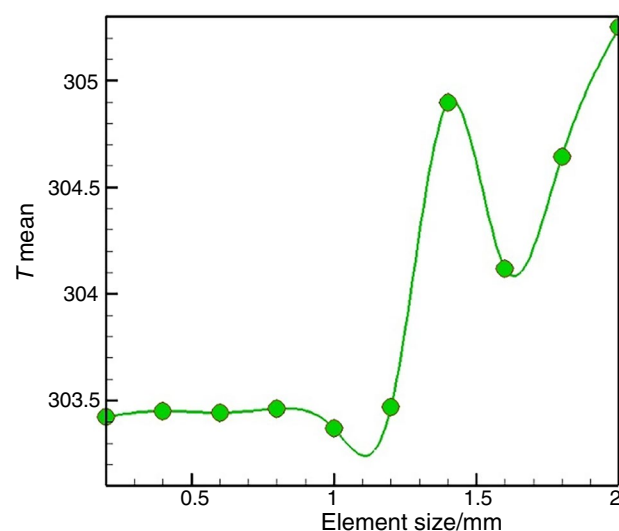


Fig. 5 Changes in the average temperature from the duct by changing the size of the rectangular elements

time will increase. Therefore, considering the independence of the answers from the element dimensions in this range, it is better to choose element dimensions close to 1 mm to reduce the analysis time as much as possible. Also, to better understand this two-dimensional model, the calculated values are presented in Table 3 for each element size and the difference from its previous value.

Validation

The present model is validated by comparison with the benchmark study reported in [51], which addresses a physically similar mixed convection heat transfer problem with vortex-generating elements. This reference was chosen because it provides a reliable basis for evaluating the accuracy of the numerical method in predicting both thermal and flow fields. While small differences in setup may lead to slight deviations in absolute values, the comparison shows consistent trends and acceptable agreement in the key quantities of interest. This level of agreement is sufficient to confirm the validity of the numerical approach adopted in the present work.

In order to verify the results, first, semicircular and triangular vortex generators are created in the channel walls according to reference [51] and the obtained results are compared with the results presented in reference [51].

The vortex generators created to verify the results are shown in Fig. 6.

The parameters related to the dimensions of the above-mentioned channel are presented in Table 4.

The contours obtained for the two geometries studied for Reynolds number 5000 are shown in Figs. 7–11. These contours are presented to qualitatively examine the distribution of various parameters.

By carefully examining the above contours, it can be seen that these contours are very similar to the contours presented in reference [51].

Figure 12 shows a comparison of the calculated outlet temperature for the channel with a triangular vortex generator in this study with the results of reference [51].

As shown in the figure above, the overall behavior of both the two-waves is qualitatively quite similar. The largest difference between the calculated outlet temperature in the present study and the reference is at low Reynolds numbers, and with increasing Reynolds number, the difference between the two graphs decreases significantly, indicating that the results of both studies are consistent.

Figure 13 demonstrate a comparison of the calculated outlet temperature for the channel with a circular vortex

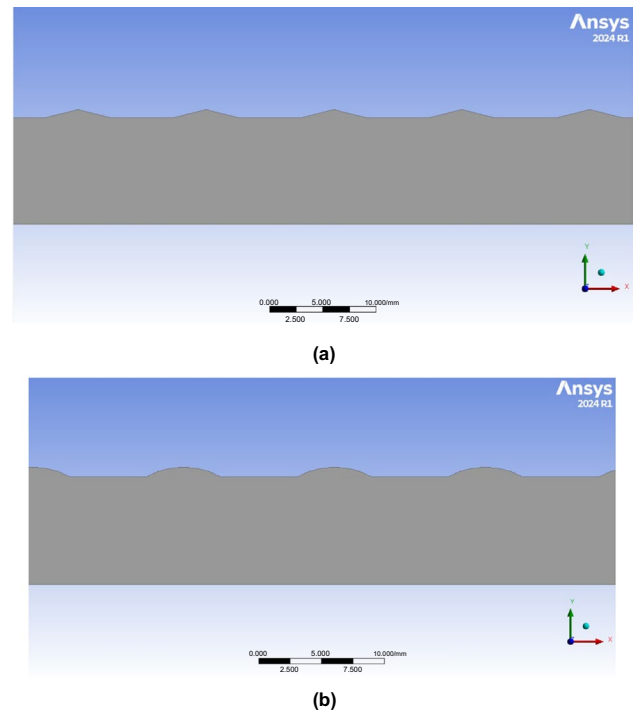


Fig. 6 Vortex generators created, a. Triangulars, b. Circulars

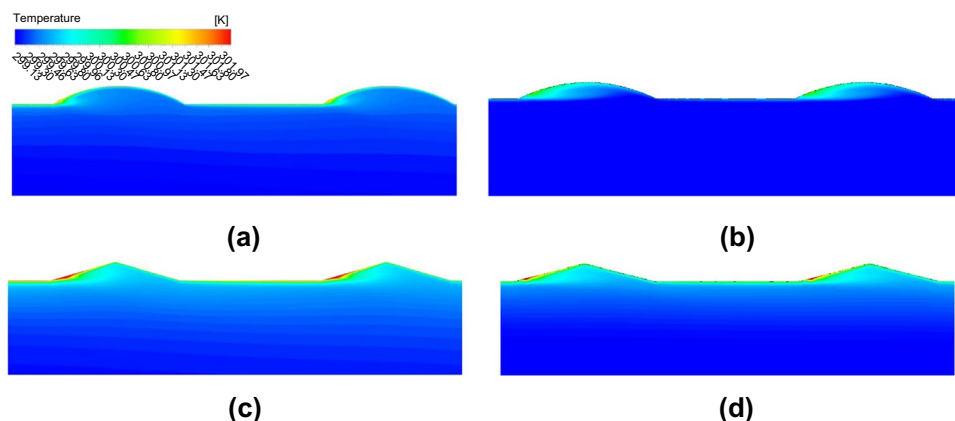
generator in this study with reference [51]. It can be seen in this figure that, like the channel with a triangular vortex generator, the difference between the two is greater at low Reynolds numbers, and with increasing Reynolds number, the difference between the two decreases. However, the difference between the two is slightly greater at high Reynolds numbers compared to the channel with a triangular vortex generator. However, the overall difference between the two graphs is still negligible and the accuracy of the results of this study can be confirmed.

Figure 14 demonstrate a comparison of the Nusselt number calculated for a channel with a triangular vortex generator in this study with reference [51].

It can also be seen in the above figure that the Nusselt number calculated in this study is very close to the Nusselt number calculated in reference [51], which confirms the high accuracy of the results obtained in this study. Figure 15 shows the comparison graphs of the changes in the Nusselt number with respect to the change in the Reynolds number in the channel with a circular vortex generator.

In this figure, it can be seen that the behavior of the two diagrams is similar to the representation of a channel with a triangular vortex generator; this means that the diagram calculated in this study and the diagram presented in the aforementioned reference have the same behavior with a small and constant difference.

Fig. 7 Temperature contours for channel with **a** circular ribs in [51], **b** circular ribs in present study, **c** triangular ribs in [51] and **d** triangular ribs in present study



Results

Heat flux boundary conditions in a wall

Figure 16 depicts the temperature distribution pattern inside the channel under the condition of applying a constant heat flux of 1000 W m^{-2} to the lower wall, in consecutive time intervals. This simulation was performed in conditions where there is no incoming or outgoing fluid flow in the system.

The results of the analysis show that with the progress of time, two important phenomena occur: first, the expansion of the heating-affected zone in the volume of fluid trapped in the channel and second, the gradual increase in the maximum temperature recorded in the system. A close examination of the temperature contours reveals that heat transfer in the central parts of the lower wall occurs with greater intensity, which has led to the creation of a larger area with high temperature in the fluid adjacent to these areas.

The dominant mechanism in these conditions is conductive heat transfer in a stationary fluid, which with the passage of time, the temperature gradient in the fluid decreases and the temperature at different points in the channel moves toward thermal equilibrium. The temperature distribution

pattern clearly shows that the highest density of isothermal lines is concentrated near the heated wall, indicating a sharp temperature gradient in this area.

Double-wall heat flux boundary conditions

Figure 17 shows the temperature distribution pattern inside the channel under the condition of applying the same heat flux of 1000 W m^{-2} to both the upper and lower walls, at successive time intervals. Examination of these contours shows that although the overall temperature distribution pattern is similar to the single-wall heated case, the main difference is in the simultaneous development of the heated zone from both the upper and lower surfaces of the channel.

The results indicate that:

- (1) The heating-affected zone extends symmetrically from both walls toward the center of the channel.
- (2) The temperature gradient near both heated walls is similar and the heat transfer intensity is comparable at both surfaces.
- (3) As time progresses, the central region of the channel, which is the last point to be affected by heating, gradually reaches a higher temperature.

Fig. 8 Velocity contours for channel with **a** circular ribs in [51], **b** circular ribs in present study, **c** triangular ribs in [51] and **d** triangular ribs in present study

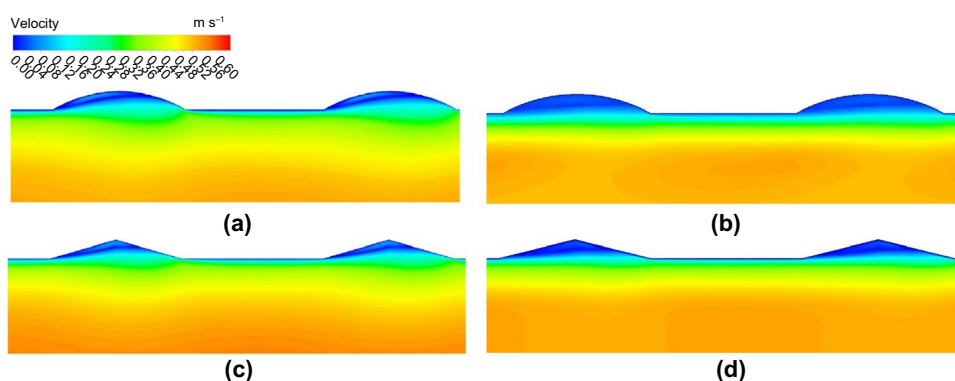


Fig. 9 Pressure contours for channel with **a** circular ribs in [51], **b** circular ribs in present study, **c** triangular ribs in [51] and **d** triangular ribs in present study

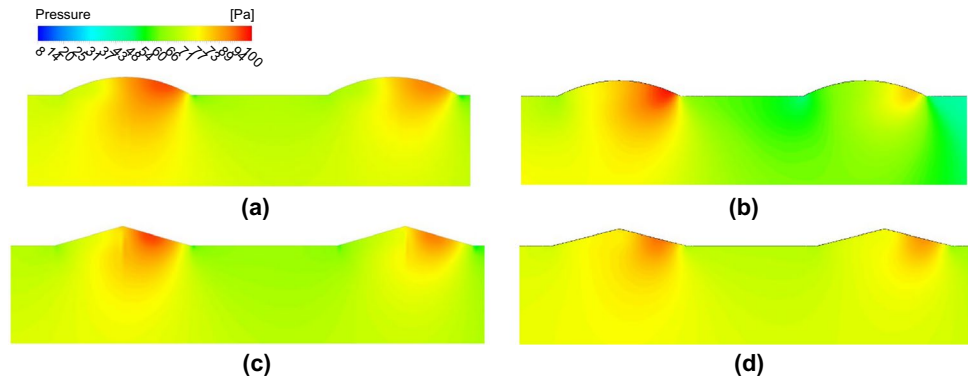
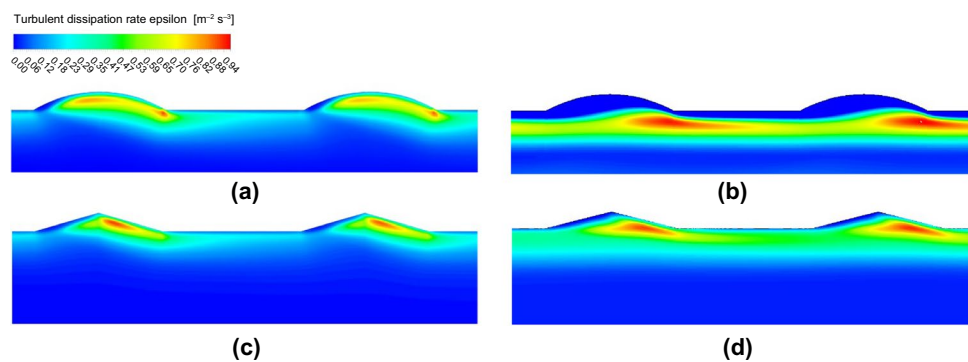


Fig. 10 Turbulence Kinetic Energy contours for channel with **a** circular ribs in [51], **b** circular ribs in present study, **c** triangular ribs in [51] and **d** triangular ribs in present study



- (4) The temperature distribution pattern shows considerable symmetry with respect to the central axis of the channel.

These conditions lead to the creation of a symmetrical temperature field with maximum temperature in the areas adjacent to both walls and minimum temperature in the central region of the channel. Over time, the temperature difference between the center and the walls decreases and the system moves toward thermal equilibrium.

Figure 18 shows the temperature changes inside the channel over time for nanofluids containing different volume

percentages of nanoparticles (1%, 2%, 3% and 4%). The results of this analysis reveal several notable points in the thermal behavior of the system:

General temperature change trend:

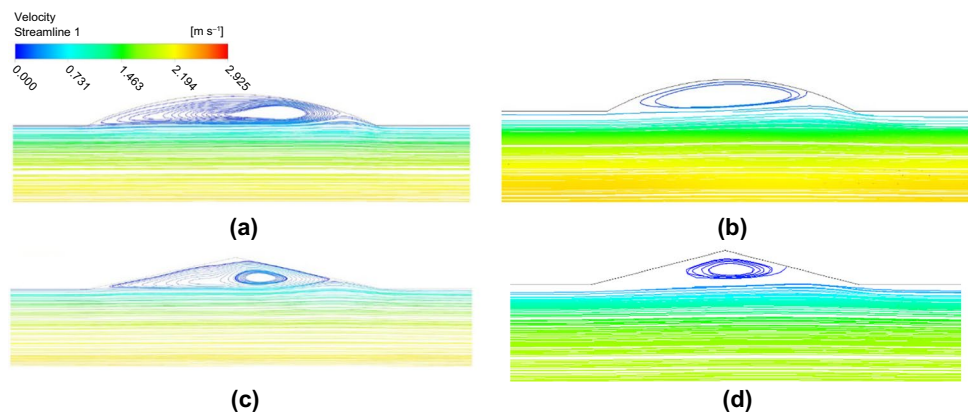
The system shows a continuous temperature increase trend in all cases.

In the time interval of 120–240 s, the system enters a transition phase and the temperature increase rate decreases clearly.

After passing through this phase, the system continues to increase its temperature again with a higher slope.

Effect of nanoparticle volume fraction:

Fig. 11 Flow lines in channel with **a** circular ribs in [51], **b** circular ribs in present study, **c** triangular ribs in [51] and **d** triangular ribs in present study



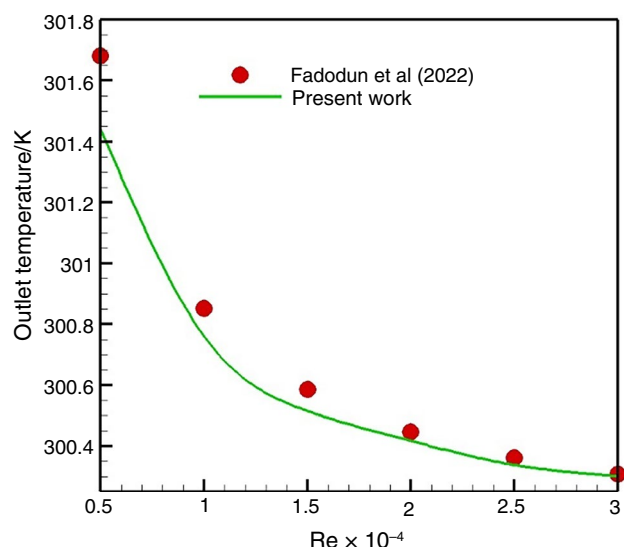


Fig. 12 Verification of outlet temperature for a channel with a triangular vortex generator with [51]

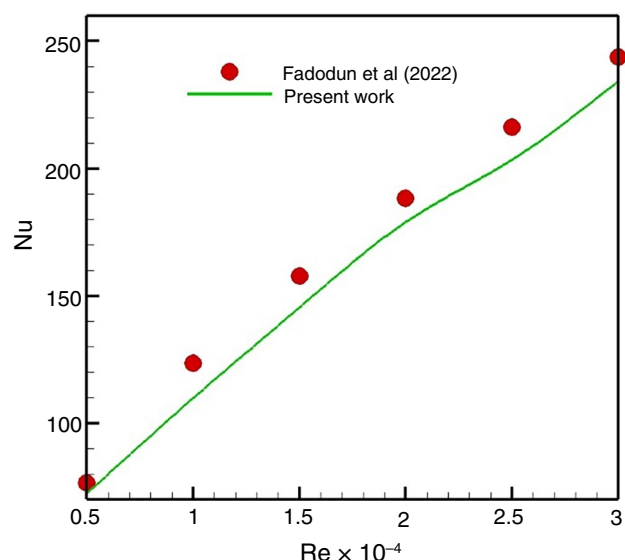


Fig. 14 Validation of Nusselt number for a channel with a triangular vortex generator

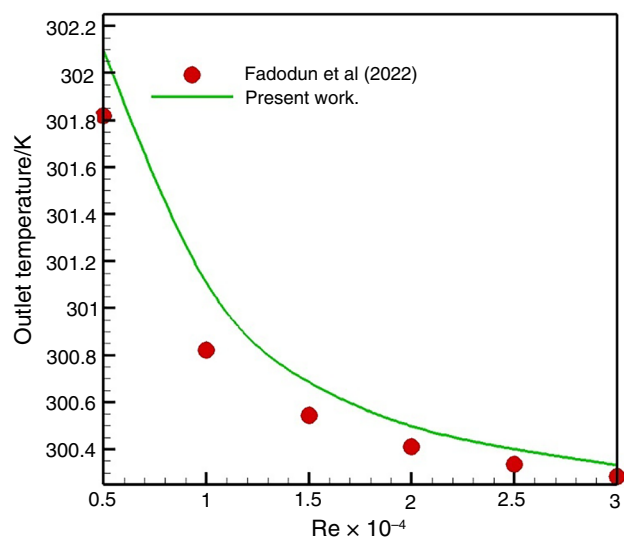


Fig. 13 Verification of outlet temperature for a channel with a circular vortex generator

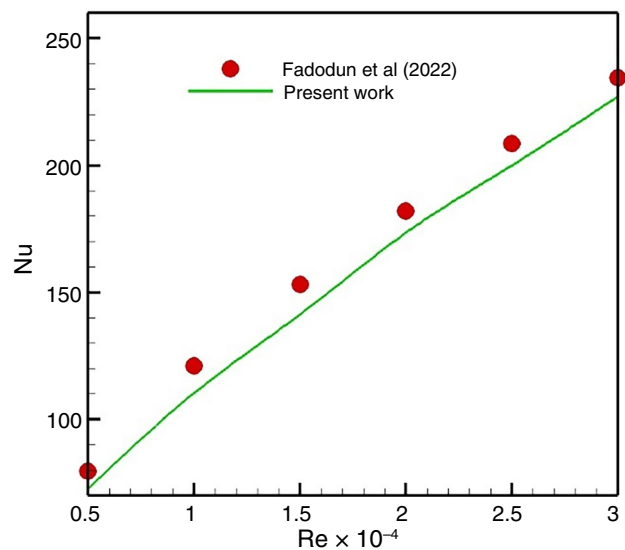


Fig. 15 Validation of Nusselt number for a channel with a circular vortex generator

Increasing the percentage of nanoparticles leads to an increase in the maximum temperature of the system.

A significant behavioral difference is observed between nanofluids with different percentages in the early stages of heating.

With the passage of time, the temperature difference between different samples gradually decreases.

Temperature convergence and saturation point phenomenon:

Over longer time scales, the plots for different percentages of nanoparticles approach each other and converge at the output.

This indicates a decreasing influence of the volume fraction of nanoparticles on the thermal behavior of the system as steady-state conditions are approached. Physically, this saturation behavior occurs due to competing

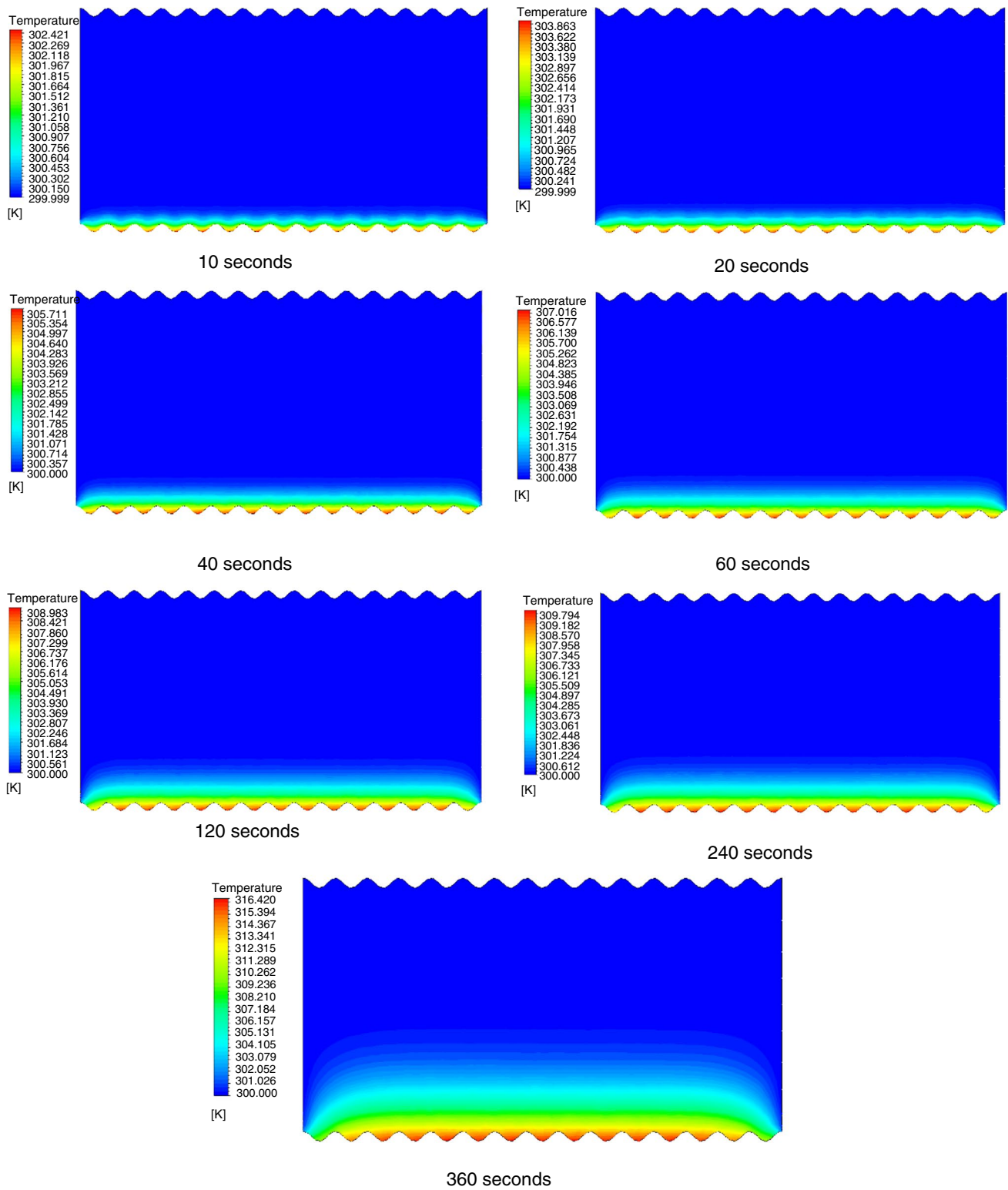


Fig. 16 Temperature distribution contours in the duct, for the lower wall case under the influence of a heat flux of 1000 W m^{-2}

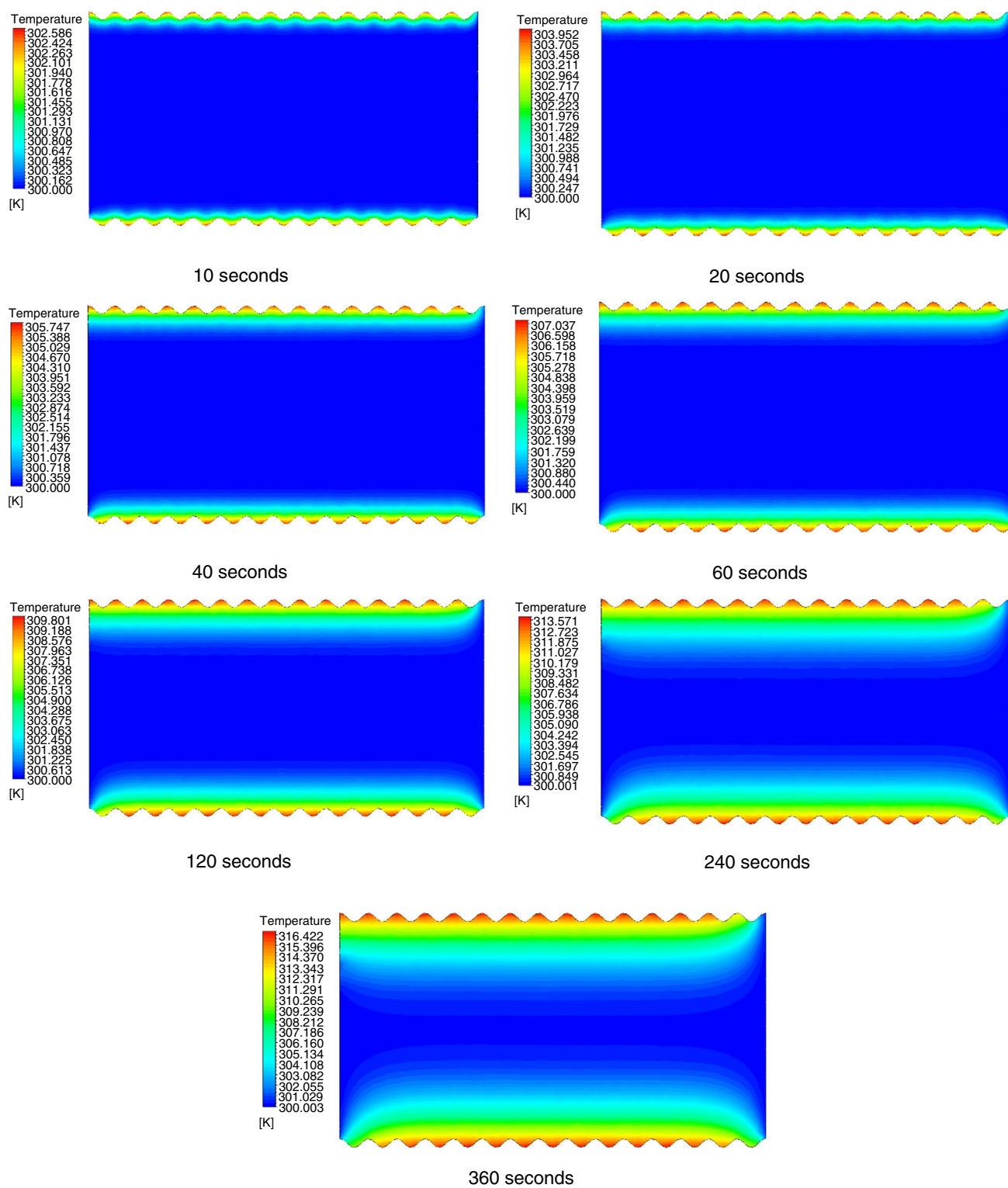


Fig. 17 Temperature distribution contours in the duct, for the upper and lower wall case under the influence of a heat flux of 1000 W m^{-2}

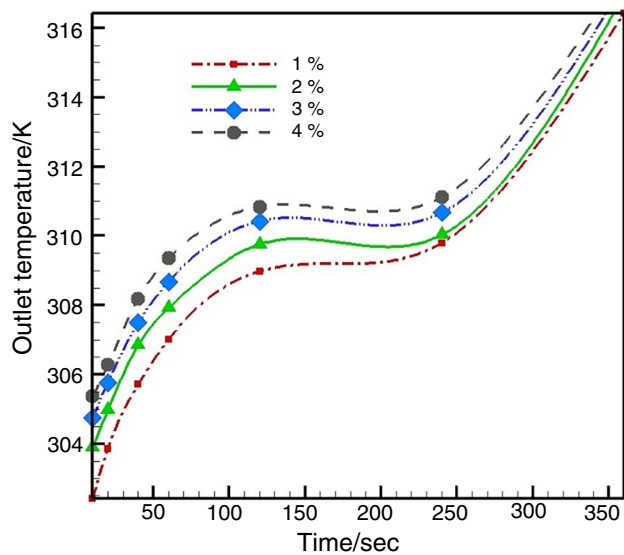


Fig. 18 Temperature changes over time

thermophysical mechanisms. While increasing the concentration of nanoparticles increases the effective thermal conductivity, it simultaneously causes a significant increase in the dynamic viscosity of the nanofluid, which suppresses fluid motion and convective heat transfer. Furthermore, over time, the system approaches thermal equilibrium, which reduces the local temperature gradients that cause convection. As a result, we reach a saturation point where adding more nanoparticles has a negligible effect on the overall thermal performance, leading to the observed temperature convergence.

Moving wall boundary condition

To investigate the effect of the moving wall velocity, velocities of 0.01, 0.05, 0.1, 0.5, 1, and 2 m s⁻¹ have been considered for the wall. To better understand the boundary conditions, Table 5 presents the Reynolds number and the governing flow regime for different moving wall velocities and different nanofluid volume fractions.

Figure 19 depicts the temperature distribution changes inside the channel when the heated wall moves at different speeds. As the wall speed gradually increases from 0.01 to 0.1 m s⁻¹, two major changes in the thermal pattern are observed: first, a significant decrease in the maximum fluid temperature, which is due to the shortening of the interaction time of the particles with the heated surface, and second, the limitation of the heating zone to the areas adjacent to the wall. At lower speeds, heat transfer penetrates deeper into the fluid volume, while at higher speeds, the heating effect remains concentrated only in the boundary layer around the wall.

As illustrated in this Fig with increasing wall velocity, the region of elevated temperature is progressively restricted to the vicinity of the moving wall. This trend can be interpreted as a relative increase in advective transport compared to thermal diffusion, resulting in a thinner thermal boundary layer and reduced heat penetration into the bulk flow,

Interestingly, the maximum temperature is concentrated at the points from which the moving wall is moving away, a phenomenon that is related to the direction of the wall movement from right to left. This asymmetric pattern is quantitatively confirmed in Fig. 20, where the inverse relationship

Table 5 Reynolds number and flow regime for moving wall

Velocity / (m s ⁻¹)	Re ($\phi=1\%$)	Flow regime	Re ($\phi=2\%$)	Flow regime	Re ($\phi=3\%$)	Flow regime	Re ($\phi=4\%$)	Flow regime
0.01	1058	Transitional	1062	Transitional	1063	Transitional	1061	Transitional
0.05	5288	Turbulent	5310	Turbulent	5314	Turbulent	5303	Turbulent
0.1	10,576	Turbulent	10,619	Turbulent	10,628	Turbulent	10,606	Turbulent
0.5	52,880	Turbulent	53,095	Turbulent	53,140	Turbulent	53,030	Turbulent
1.0	105,760	Turbulent	106,190	Turbulent	106,280	Turbulent	106,060	Turbulent
2.0	211,520	Turbulent	212,380	Turbulent	212,560	Turbulent	212,120	Turbulent

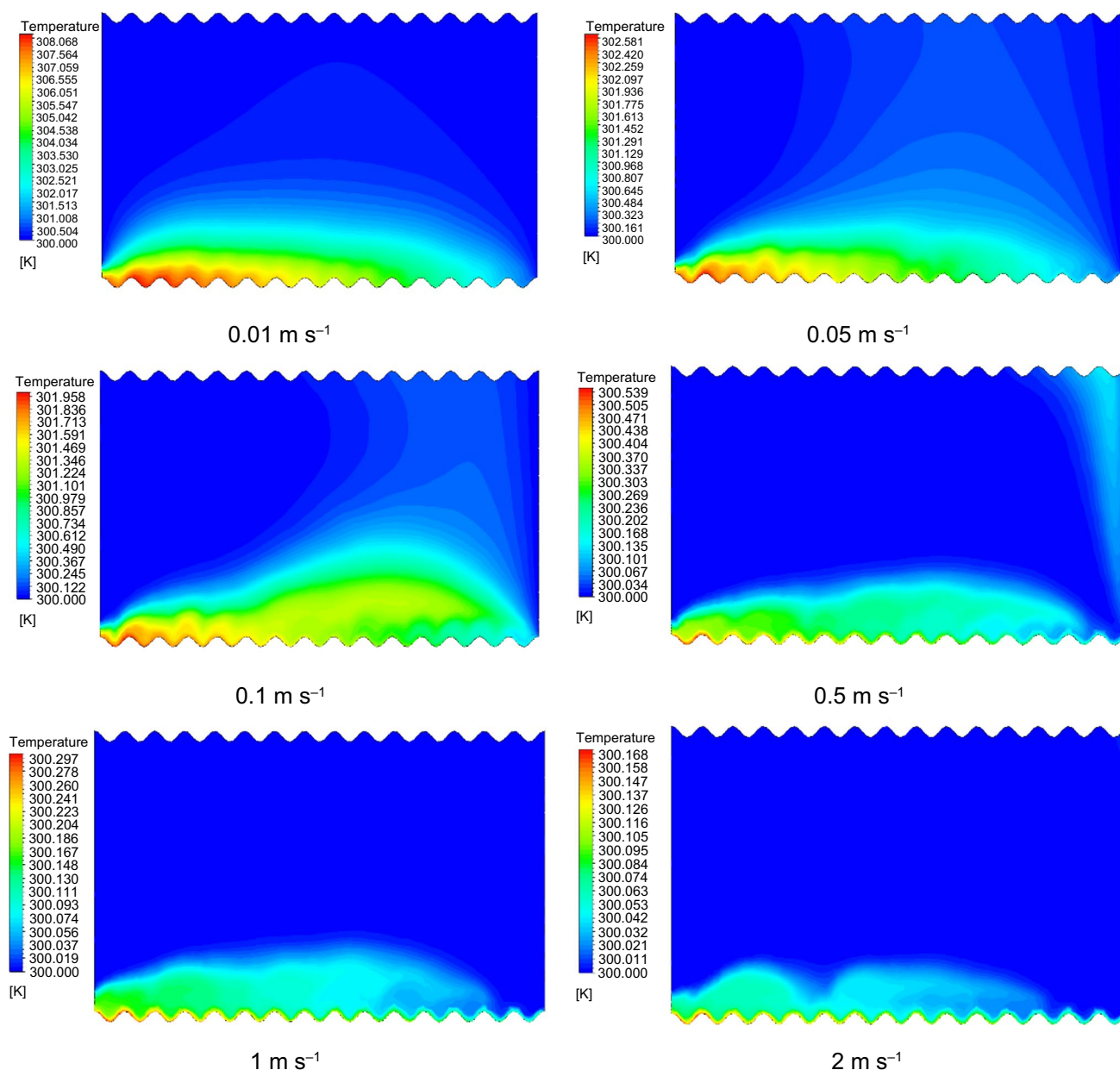


Fig. 19 Temperature distribution contour in the duct when the lower wall has heat flux and is moving

between the wall speed and the maximum fluid temperature is clearly visible. As the velocity increases, the maximum temperature continues to decrease with a nearly constant slope, indicating a shift in the balance between conduction and convection mechanisms in the heat transfer of the system.

Analysis of the results presented in the figure above shows that the changes in the maximum temperature of the fluid in response to the increase in the speed of the moving wall follow a specific pattern. In the first stage, when the wall speed increases from zero to 0.1 m s^{-1} , we witness a sharp and sudden decrease in the maximum temperature,

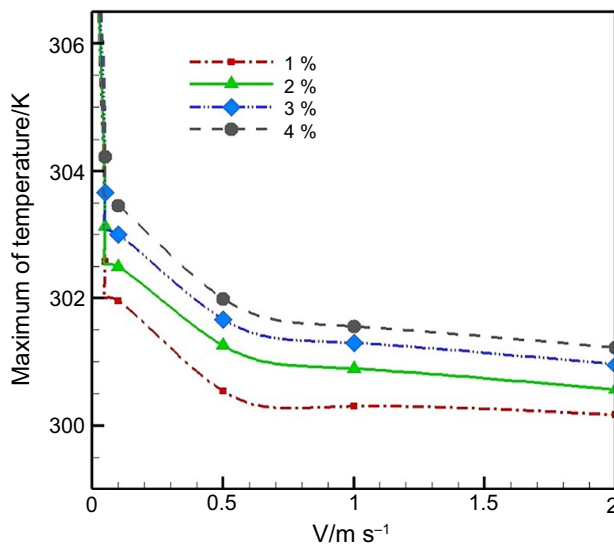


Fig. 20 Maximum temperature changes relative to the moving wall speed

which is observed for all volume percentages of nanoparticles in the same way. This phenomenon indicates the high sensitivity of the system to changes in speed in the low-speed range. With a further increase in speed and passing the threshold of 0.1 m s^{-1} , the temperature change process enters a new phase in which the rate of temperature decrease gradually decreases. At this stage, although the increase in speed still causes the maximum temperature to decrease, the intensity of this decrease is much less than in the previous stage. This behavior continues up to a speed of about 0.5 m s^{-1} . At speeds higher than 0.5 m s^{-1} , the temperature graph approaches a nearly horizontal state, indicating that the system has reached equilibrium and saturation. In this situation, further increase in velocity has no noticeable effect on the maximum fluid temperature, because the heat transfer mechanisms have reached their maximum efficiency and the contact time of the fluid particles with the wall has been reduced to a minimum. These results clearly show that the system is most sensitive to velocity changes in the low velocity range (below 0.1 m s^{-1}), while it becomes almost indifferent at high velocities (above 0.5 m s^{-1}).

Fixed wall and flowing fluid in the channel

Figure 21 shows that as the mass flow rate of the inlet nanofluid increases, the maximum temperature inside the channel decreases. Also, the region where the temperature of the nanofluid increases becomes smaller and more limited. In the inlet section of the channel, the maximum temperature and the heating area are more limited, while in the outlet section these areas are wider.

Also, Table 6 presents the Reynolds number and flow regime for different flow rates and volume fractions of this case.

Figure 22 shows the amount of heat absorbed by the nanofluid for different nanoparticle concentrations and different mass flow rates. It can be seen that when the mass flow rate increases from 0.001 to 0.1 kg s^{-1} , the heat transfer increases sharply, but after this point, the growth of the graph slows down noticeably. However, the overall trend is still upward, although at a lower rate. It is also clear that increasing the percentage of nanoparticles in the nanofluid improves heat transfer, but this effect gradually weakens with increasing nanoparticle concentration.

The initial enhancement in heat transfer with increasing nanoparticle concentration can be attributed to the increase in effective thermal conductivity. However, at higher concentrations, the increase in effective viscosity becomes more dominant, which raises flow resistance, thickens the thermal and velocity boundary layers, and weakens fluid mixing. Consequently, the heat transfer improvement gradually diminishes and tends to saturate beyond a certain nanoparticle concentration.

Also, according to the behavior of the graphs in the above figure, the initial increase in heat transfer with increasing nanoparticle concentration can be attributed to the increase in effective thermal conductivity. However, at higher concentrations, the increase in effective viscosity becomes more dominant, which increases the flow resistance, thickens the thermal boundary layers and the velocity and weakens the fluid mixing. As a result, the heat transfer improvement gradually decreases and tends to saturate beyond a certain nanoparticle concentration.

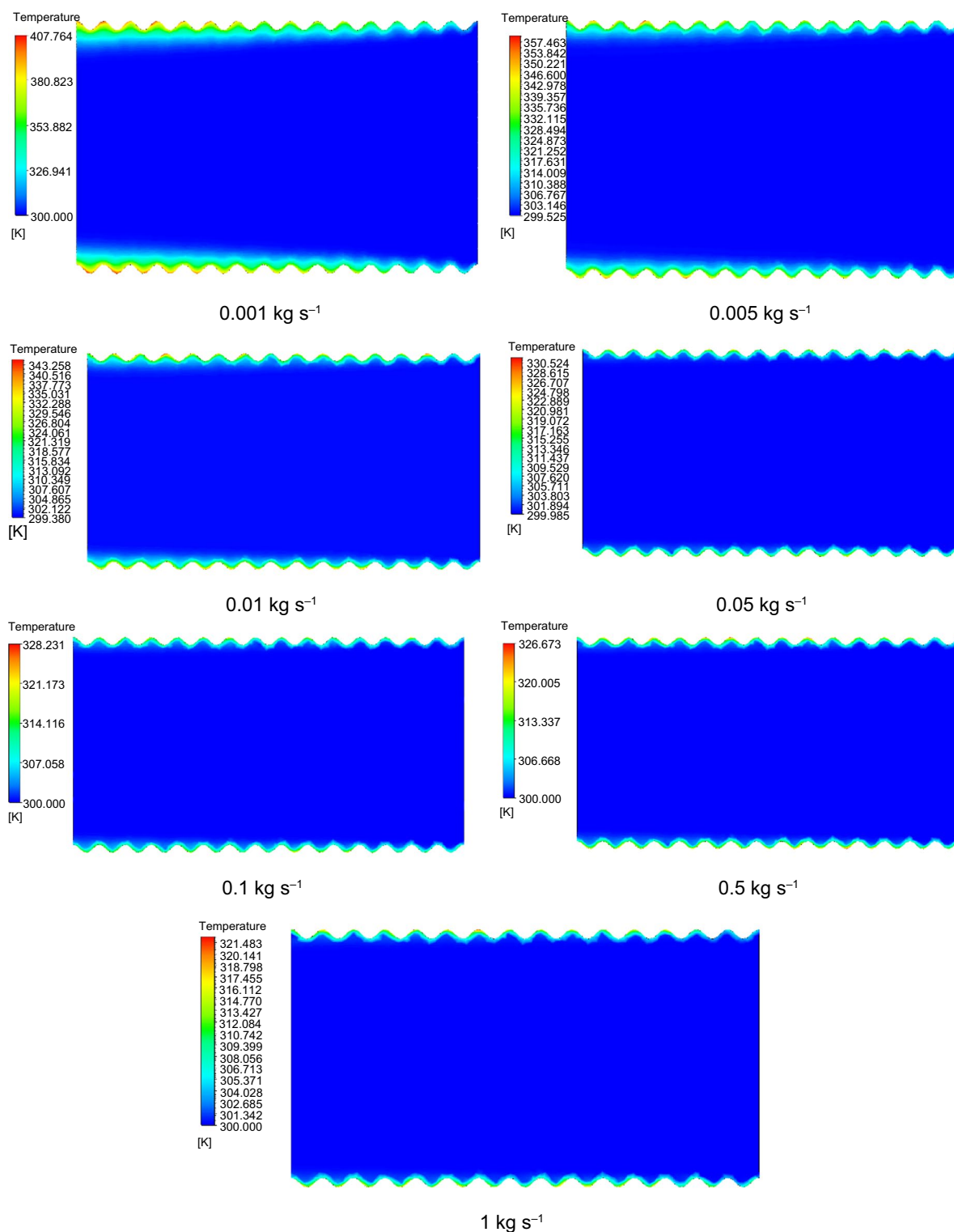
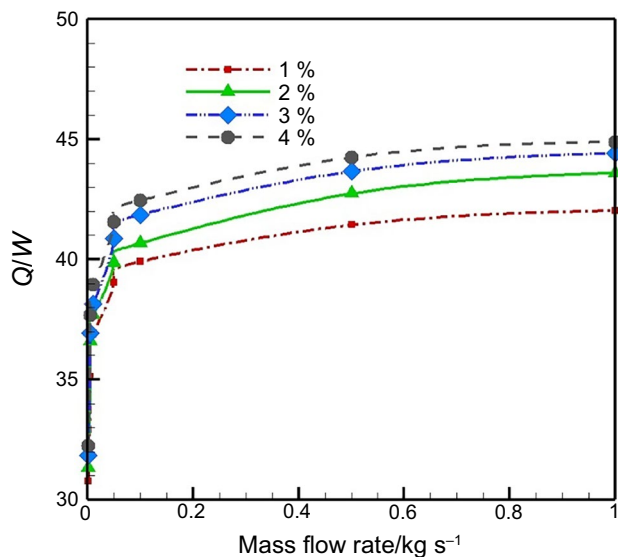


Fig. 21 Temperature distribution contour in the duct when the upper and lower wall has heat flux and is moving

Table 6 Reynolds number and flow regime for fixed wall and flowing fluid

Mass flow rate / (kg s ⁻¹)	Re ($\phi=1\%$)	Flow regime	Re ($\phi=2\%$)	Flow regime	Re ($\phi=3\%$)	Flow regime	Re ($\phi=4\%$)	Flow regime
0.001	2.04	Laminar	2.00	Laminar	1.97	Laminar	1.96	Laminar
0.005	10.2	Laminar	10.0	Laminar	9.85	Laminar	9.80	Laminar
0.01	20.4	Laminar	20.0	Laminar	19.7	Laminar	19.6	Laminar
0.05	102	Laminar	100	Laminar	98.5	Laminar	98.0	Laminar
0.1	204	Laminar	200	Laminar	197	Laminar	196	Laminar
0.5	1020	Laminar	1000	Laminar	985	Laminar	980	Laminar
1.0	2040	Laminar	2000	Laminar	1970	Laminar	1960	Laminar
5.0	10,200	Turbulent	10,000	Turbulent	9,850	Turbulent	9,800	Turbulent

**Fig. 22** Heat absorbed by nanofluids with different percentages of nanoparticles for different mass flow rates

Conclusions

In this study, the mixed convection heat transfer of a water-Fe₃O₄ nanofluid in a corrugated channel with moving walls was numerically investigated under both steady and transient conditions. The key findings are summarized as follows:

- The thermal behavior of the fluid is significantly influenced by nanoparticle concentration, wall velocity, and time.
- Increasing the nanoparticle volume fraction initially increases the maximum temperature of the fluid; however, this effect diminishes over time as the system approaches thermal equilibrium.
- In the moving wall scenario, increasing the wall speed effectively lowers the maximum fluid temperature at lower speeds. Nevertheless, this cooling effect saturates

at speeds beyond 0.5 m s⁻¹, where further increases in velocity yield negligible changes in temperature.

- Overall, the study demonstrates that heat transfer and thermal management in corrugated channels can be effectively controlled by optimizing both the nanoparticle concentration and the wall motion.

Author contribution N.F. contributed to conceptualization, investigation, derivation, formulation, software, validation, supervision, and writing—original draft. A.I.R. contributed to investigation, formal analysis, data curation, visualization, writing—review and editing. J.M.A. contributed to conceptualization, investigation, resources, data analysis, and writing—original draft. A.K.A. contributed to investigation, resources, data curation, visualization, writing—review and editing. All authors reviewed the manuscript.

Funding Open Access funding enabled and organized by Projekt DEAL.

Data availability Data will be available on request from the corresponding author.

Declarations

Conflict of interest The authors know no conflict of interest to declare.

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